A BURGERS–FISHER EQUATION FOR PRIME GAPS: SPECTRAL ANALYSIS OF CONVERGENCE TO POISSON

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Abstract. We propose a PDE model for the convergence of prime gaps to the Poisson regime. If the density f of normalized prime gaps $v = g/\log x$ evolves under the Feller Fokker–Planck equation —chosen because its stationary distribution is Exp(1) and its state-dependent diffusivity matches the empirical rigidity of small gaps—then the velocity field $u = -\partial_v \log f$ satisfies a Burgers–Fisher equation in a porous medium:

$$\partial_t u + 2v\, u\, \partial_v u = v\, \partial_v^2 u + (v{+}2)\, \partial_v u + u(1{-}u).$$

The Fisher–KPP reaction term $u(1-u)$ emerges from the Fokker–Planck structure and drives u toward 1 (the Poisson equilibrium). The linearized perturbation operator has associated Laguerre polynomial eigenfunctions with spectral gap 1, and perturbations with decaying forcing vanish as $O(1/(\log x)^\beta)$. Numerical verification at 11 scales from 10^3 to 10^8 shows the mutual information between consecutive gaps decays as $(\log x)^{-0.19}$ ($R^2 = 0.986$) and that prime $p = 3$ carries 94% of the sieve signal.

Contents

1. Introduction 2
2. The Burgers–Fisher Equation for Prime Gaps 2
2.1. The Feller process and its Fokker–Planck equation 2
2.2. Derivation of the velocity equation 2
3. Spectral Stability of the Poisson Equilibrium 3
4. Convergence Under Decaying Sieve Forcing 4
5. Connection to Prime Gaps 4
5.1. The sieve as porous medium 4
5.2. Exact computation: the mod-3 coupling 5
6. Numerical Verification 5
6.1. MI decay rate 5
6.2. Prime 3 dominance 5
6.3. Perturbation energy 5
6.4. Information cascade 5
6.5. Sieve diffusion model 5
7. Discussion: What the Paper Does and Does Not Prove 6
7.1. The role of the PDE model 6
7.2. Open problems 6
Computational Details 6
Use of generative AI 7

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References 7

1. Introduction

Gallagher [1] proved, conditionally on the Hardy–Littlewood conjecture [2], that prime counts in short intervals converge to Poisson statistics. Lemke Oliver and Soundararajan [3] discovered biases in consecutive prime residues decaying as $O((\log\log x)/\log x)$; Holt [4] modeled these biases via sieve dynamics. However, no quantitative convergence rate for the overall gap distribution to Poisson has been established.

We construct a velocity field from the prime gap distribution and show it satisfies a nonlinear PDE whose unique stable equilibrium is the Poisson regime. This provides both a mechanism (the spectral gap of a Laguerre-type operator) and, numerically, a rate ($\approx (\log x)^{-0.19}$).

2. The Burgers–Fisher Equation for Prime Gaps

2.1. The Feller process and its Fokker–Planck equation.

We seek a diffusion process whose stationary distribution is Exp(1)—the Gallagher–Poisson target. Among processes with linear drift, the unique choice whose diffusion coefficient also matches the empirical structure of prime gaps is the Feller diffusion: its noise $\sigma(V) = \sqrt{2V}$ means small gaps ($V \approx 0$) are nearly deterministic while large gaps fluctuate freely, reflecting the tight arithmetic constraints the sieve imposes on small gaps (only $g = 2, 4, 6$ are common) versus the broad distribution of large gaps. Cohen [7] independently confirmed that the moments of prime gaps converge to those of the exponential distribution, further supporting this choice.

Consider the Feller diffusion (Cox–Ingersoll–Ross process)

$$dV = (1 - V)\,dt + \sqrt{2V}\,dW_t, \qquad V \geq 0, \tag{1}$$

with mean-reversion toward 1 and state-dependent noise $\sigma(V) = \sqrt{2V}$. Its Fokker–Planck equation for the density $f(v, t)$ is

$$\partial_t f = v f'' + (v + 1) f' + f, \tag{2}$$

where primes denote ∂_v.

Lemma 2.1. *$f_\infty(v) = e^{-v}$ is the unique probability density satisfying $\mathcal{L} f_\infty = 0$, where $\mathcal{L} f = v f'' + (v + 1) f' + f$.*

Proof. Direct verification: $\mathcal{L}(e^{-v}) = v \cdot e^{-v} + (v + 1)(-e^{-v}) + e^{-v} = 0$. Uniqueness follows from the general theory of Feller diffusions [6]: the process (1) with $2 \times$ mean-reversion speed$/\sigma^2 = 2 \cdot 1/2 = 1 \geq 1$ has an entrance boundary at 0 and a natural boundary at ∞, admitting a unique stationary distribution. □

2.2. Derivation of the velocity equation.

Theorem 2.2 (The sieve Burgers–Fisher equation)**.** *Let $f(v, t) > 0$ satisfy the Fokker–Planck equation* (2) *on* $(0, \infty)$*. Define the velocity field*

$$u(v, t) = -\frac{\partial}{\partial v} \log f(v, t) = -\frac{f'}{f}. \tag{3}$$

Then u satisfies the Burgers–Fisher equation in a porous medium:

$$\boxed{\partial_t u + 2v\, u\, \partial_v u = v\, \partial_v^2 u + (v{+}2)\, \partial_v u + u(1{-}u).} \tag{4}$$

Proof. Set $\psi = \log f$, so $\psi' = f'/f$ and $f''/f = \psi'' + (\psi')^2$. From (2):

$$\psi_t = v(\psi'' + \psi'^2) + (v+1)\psi' + 1. \tag{5}$$

Differentiate (5) with respect to v:

$$\begin{aligned}\psi_t' &= \psi'' + v\psi''' + \psi'^2 + 2v\psi'\psi'' + \psi' + (v+1)\psi'' \\ &= v\psi''' + (v+2)\psi'' + 2v\psi'\psi'' + \psi'^2 + \psi'.\end{aligned}$$

Substitute $u = -\psi'$ (hence $\psi'' = -u'$, $\psi''' = -u''$):

$$-u_t = -vu'' - (v+2)u' + 2vu\,u' + u^2 - u.$$

Multiply by -1 and rearrange:

$$u_t = vu'' + (v{+}2)u' - 2vu\,u' - u^2 + u = vu'' + (v{+}2)u' + u(1{-}u) - 2vu\,u'. \quad \square$$

Remark 2.3 (Structure of the equation)**.** Equation (4) combines three classical PDE mechanisms:

(1) *Burgers nonlinearity* $2vu\,u_v$: the hallmark of the Navier–Stokes and Burgers equations, arising here from the nonlinear transformation $u = -(\log f)'$.
(2) *Porous medium diffusion* $v\,u_{vv}$: the diffusivity equals v, so small gaps ($v \approx 0$) are "tight" and large gaps fluctuate freely.
(3) *Fisher–KPP reaction* $u(1{-}u)$: this term pushes u toward 1 from either side. It emerges from the Fokker–Planck structure, not imposed by hand.

3. Spectral Stability of the Poisson Equilibrium

Theorem 3.1 (Spectral analysis)**.**

(i) *The unique bounded steady state of* (4) *is* $u \equiv 1$*, corresponding to the exponential distribution* $f = e^{-v}$*.*
(ii) *The linearization around* $u = 1$*, writing* $\delta u = u - 1$*, is*

$$\partial_t(\delta u) = v\,(\delta u)'' + (2-v)\,(\delta u)' - \delta u. \tag{6}$$

(iii) *The eigenvalue problem* $v\,h'' + (2-v)\,h' + (\lambda+1)\,h = 0$ *is the associated Laguerre equation with parameter* $\alpha = 1$*. Its solutions are the associated Laguerre polynomials* $L_n^{(1)}(v)$ *with eigenvalues*

$$\lambda_n = -(n+1), \qquad n = 0, 1, 2, \ldots \tag{7}$$

(iv) *The spectral gap is* $|\lambda_0| = 1$*.*

Proof. (i) A constant steady state $u = c$ satisfies $c(1-c) = 0$, giving $c = 0$ or $c = 1$. If $u = 0$ then $f' = 0$, i.e., $f = \text{const}$, which is not integrable on $[0, \infty)$. Thus $u = 1$ is the unique physically admissible steady state.

(ii) Substitute $u = 1 + \delta u$ into (4), keeping linear terms in δu:

$$\begin{aligned}(\delta u)_t + 2v \cdot 1 \cdot (\delta u)' &= v(\delta u)'' + (v+2)(\delta u)' + (1+\delta u)(1-1-\delta u) \\ &= v(\delta u)'' + (v+2)(\delta u)' - \delta u + O((\delta u)^2).\end{aligned}$$

Hence $(\delta u)_t = v(\delta u)'' + (2-v)(\delta u)' - \delta u$.

(iii) Setting $(\delta u)(v,t) = h(v)\,e^{\lambda t}$ gives $vh'' + (2-v)h' + (\lambda+1)h = 0$. With the substitution $\alpha = 1$ and $n = \lambda + 1$, this is the associated Laguerre equation $vh'' + (\alpha + 1 - v)h' + nh = 0$, whose polynomial solutions are $L_n^{(1)}(v)$ [5].

(iv) From $\lambda_n = -(n+1)$: $\lambda_0 = -1$, so all perturbation modes decay at rate ≥ 1. □

4. Convergence Under Decaying Sieve Forcing

Theorem 4.1 (Perturbed convergence)**.** *Let $u(v,t)$ satisfy*

$$\partial_t u + 2vu\,u_v = vu_{vv} + (v+2)u_v + u(1-u) + \varepsilon(t)\,F(v), \tag{8}$$

where $\|F\|_\infty \leq M$ and $|\varepsilon(t)| \leq C_0/t^\beta$ for $\beta > 0$. Then there exists $C > 0$ such that $\|\delta u(\cdot,t)\|_{L^2(v\,e^{-v})} \leq C/t^\beta$ for all t sufficiently large.

Proof. Define the energy $E(t) = \frac{1}{2}\|\delta u\|^2_{L^2(ve^{-v})}$, where $L^2(ve^{-v})$ is the natural weighted space for the associated Laguerre polynomials.

Step 1 (Dissipation). By the spectral gap (Theorem 3.1(iv)): $\langle \mathcal{L}_{\text{lin}}(\delta u), \delta u\rangle \leq -|\lambda_0| \cdot \|\delta u\|^2 = -\|\delta u\|^2$.

Step 2 (Forcing). $|\langle \varepsilon F, \delta u\rangle| \leq |\varepsilon| \cdot M' \cdot \|\delta u\|$, where $M' = \|F\|_{L^2(ve^{-v})}$. By Young's inequality: $|\varepsilon| M' \|\delta u\| \leq \frac{1}{2}\|\delta u\|^2 + \frac{M'^2}{2}\varepsilon^2$.

Step 3 (Energy inequality). $\dot{E} \leq -2E + E + \frac{M'^2}{2}\varepsilon^2 = -E + \frac{C_1}{t^{2\beta}}$.

Step 4 (Grönwall). The integrating factor e^t gives $E(t) \leq E(t_0)e^{-(t-t_0)} + C_1 \int_{t_0}^t e^{-(t-\tau)}\tau^{-2\beta}\,d\tau$. The integral is $O(t^{-2\beta})$ by repeated integration by parts, yielding $E(t) = O(t^{-2\beta})$. □

5. Connection to Prime Gaps

Hypothesis 5.1 (Sieve diffusion)**.** Under the Hardy–Littlewood conjecture, the normalized gap density $f(v,s)$ at scale $s = \log x$ can be written as $f = e^{-v}[1 + \varphi(v,s)]$, where the perturbation φ is driven by the singular series and satisfies $\|\varphi(\cdot,s)\| = O(1/s^\beta)$ for some $\beta > 0$.

Under this hypothesis, the velocity field $u = 1 - \varphi'/(1+\varphi) \approx 1 - \varphi'$ satisfies a forced version of (4), and Theorem 4.1 gives $u \to 1$, i.e., convergence to Poisson.

Remark 5.2 (What the hypothesis does not give)**.** Hypothesis 5.1 concerns the marginal gap density. The mutual information $\text{MI}(g_n; g_{n+1})$—a joint quantity—requires additional input. We verify the MI decay numerically (§6) but leave the rigorous joint implication as an open problem.

5.1. The sieve as porous medium.

Under the Hardy–Littlewood conjecture [2], the pair singular series gives: $f(v,s) \propto \mathfrak{S}(\{0, vs\}) \cdot e^{-v}$, where the relative weight is $W(g) = \mathfrak{S}(\{0,g\})/(2C_2) = \prod_{p|g,\,p>2}(p-1)/(p-2)$. Each prime p is a "pore" with strength $\delta_p = 1/(p-2)$: $\delta_3 = 1$, $\delta_5 = 1/3$, $\delta_7 = 1/5$. As $s \to \infty$, gap values $g = vs$ grow and the mod-p structure equidistributes, weakening each pore.

5.2. **Exact computation: the mod-3 coupling.**

Proposition 5.3. *Under i.i.d. uniform prime residues mod* 3*:* $\mathrm{MI}(g_n \bmod 3;\, g_{n+1} \bmod 3) = 1/4$ *bit, with structural zeros at* $(1,1)$ *and* $(2,2)$ *in the joint table.*

Proof. With $r_n = p_n \bmod 3 \in \{1,2\}$ i.i.d. uniform, the gap $g_n \equiv r_{n+1} - r_n \pmod 3$. Enumerating all 8 triples $(r_n, r_{n+1}, r_{n+2}) \in \{1,2\}^3$ yields the joint distribution:

$g_n \bmod 3 \setminus g_{n+1} \bmod 3$	0	1	2
0	2/8	1/8	1/8
1	1/8	**0**	1/8
2	1/8	1/8	**0**

Marginals: $P(0) = 1/2$, $P(1) = P(2) = 1/4$. Only entries $(1,2)$ and $(2,1)$ deviate from the product of marginals: $\mathrm{MI} = 2 \times \frac{1}{8}\log_2(\frac{1/8}{1/4 \cdot 1/4}) = \frac{1}{4}$ bit. □

6. Numerical Verification

All measurements use $N = 200{,}000$ primes at 11 scales (10^3 to 10^8).

6.1. **MI decay rate.** $\mathrm{MI}_{\mathrm{total}}(x) \sim 0.555/(\log x)^{0.186}$, $R^2 = 0.986$. To our knowledge, this is the first numerical measurement of the MI decay rate for prime gaps.

6.2. **Prime 3 dominance.** $\mathrm{MI}_3 / \mathrm{MI}_{\mathrm{total}} \approx 94\%$ at all scales. Empirical $\mathrm{MI}_3 = 0.302$ bits vs. theoretical 0.250 bits; the 21% excess over the i.i.d. prediction is consistent with the known biases in consecutive prime residues documented by Lemke Oliver and Soundararajan [3]. The structural zeros $P(1,1) = P(2,2) = 0$ hold *exactly* for all primes > 3: if $g_n \equiv 1 \pmod 3$ then $p_{n+1} \equiv 2 \pmod 3$, forcing $g_{n+1} \equiv 1$ to require $p_{n+2} \equiv 0 \pmod 3$, which is impossible for $p_{n+2} > 3$. Verified empirically with zero exceptions in 78,493 consecutive gap pairs.

6.3. **Perturbation energy.** $E(s) = \|\varphi\|^2_{L^2(e^v)} \sim C/(\log x)^{0.63}$, $R^2 = 0.67$. The Laguerre decomposition shows the $n = 2$ mode (variance correction) dominates, while the $n = 1$ mode (mean shift) is nearly zero.

6.4. **Information cascade.**

Prime p	Decay exponent α_p	R^2	MI at 10^8
3	0.22	0.980	0.302 bits
7	3.83	0.996	0.006 bits
11	3.67	0.998	0.032 bits

Prime 3 ($\alpha \approx 0.22$) is the bottleneck; primes ≥ 7 decay as $(\log x)^{-3.5}$ or faster.

6.5. **Sieve diffusion model.** Starting from $f_0 = e^{-v}$ (Cramér) and iteratively adding sieve primes: adding $p = 3$ alone reduces KL divergence from the empirical distribution by 74% (from 0.061 to 0.016 nats at scale 10^7).

7. Discussion: What the Paper Does and Does Not Prove

7.1. **The role of the PDE model.** Gallagher [1] already proved (conditionally) that prime gaps converge to Poisson. Our paper does not give a new proof of this fact. What the paper provides is a *mechanism*: a specific PDE (the Burgers–Fisher equation) whose spectral structure explains *why* the convergence occurs and *how fast* it proceeds.

As Tao [9] observes, probabilistic models in number theory are "largely informal and fuzzy in nature" yet "indirectly enable rigorous progress" by serving as "directional guides for fruitful conjectures" and "error-detection tools." Our paper follows the pattern established by the random matrix theory program for L-functions: Keating and Snaith [8] derived the moments conjecture for $\zeta(1/2 + it)$ by modeling zeta with random matrix characteristic polynomials—a heuristic that generates precise, testable predictions while awaiting rigorous derivation. Similarly, we derive the Burgers–Fisher equation from a specific Fokker–Planck model, generate a convergence rate prediction ($\alpha \approx 0.19$), and verify it numerically, while the derivation of the model from the Hardy–Littlewood conjecture remains open.

The logical structure is:

(1) **Pure PDE (Theorems 2.2–4.1):** If a density evolves under the Feller Fokker–Planck equation, the velocity field satisfies the Burgers–Fisher equation with stable equilibrium $u = 1$ and spectral gap 1. Perturbations with decaying forcing vanish. *These are unconditional theorems.*
(2) **Model (Hypothesis 5.1):** The prime gap density at scale $s = \log x$ is well-approximated by the Feller stationary distribution perturbed by the singular series, with perturbation decaying as $O(1/s^\beta)$. *This is a modeling assumption, supported by numerical evidence but not derived from number theory.*
(3) **Consequence:** Under the model, the spectral gap drives convergence to Poisson at a computable rate.

The open challenge is to *derive* Hypothesis 5.1 from the Hardy–Littlewood conjecture, which would close the gap between the PDE theorems and number theory—as Talagrand's proof of the Parisi formula [10] closed a 27-year gap between the physics prediction and rigorous mathematics.

7.2. **Open problems.**

(1) **Derive the sieve diffusion hypothesis** from Hardy–Littlewood and equidistribution of primes in arithmetic progressions.
(2) **Prove MI** $\to 0$ from the Burgers–Fisher framework (requires extending from marginal to joint gap densities).
(3) **Compute the exponent** $\alpha \approx 0.19$ from first principles, starting from the pore strength $\delta_3 = 1$ and the equidistribution rate of gaps mod 3.

Computational Details

Python 3 with SymPy, NumPy, SciPy. Runtime ≈ 2 minutes. Code, data, and interactive visualization: `https://github.com/tobiascanavesi/prime-fluid-dynamics`

Use of generative AI. Claude Opus 4.6 (Anthropic, 2025) assisted with code development and LaTeX formatting. All mathematical content is the original contribution of the author.

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